A Study on the Effect of Dimensionality Reduction on Image Classification

ISyE 6740 - Computational Data Analytics - Spring 2023

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1 Problem Statement

Dimension reduction is a crucial technique in data science, playing a fundamental role in data visualization and as a preprocessing step for machine learning. One key benefit is the ability to shorten training and testing time for classifiers, as high-dimensional data can pose challenges in terms of computational complexity. Techniques such as Principal Component Analysis (PCA), isometric mapping (ISOMAP), uniform manifold approximation and projection (UMAP) can significantly accelerate the training and testing process of machine learning algorithms, making them more efficient and practical for real-world applications. High-dimensional datasets often contain irrelevant or redundant information and impact the performance of machine learning models. Dimension reduction methods can identify and eliminate noise information, mitigating the risk of overfitting, leading to better predictive performance, and increased model interpretability.

Furthermore, dimension reduction can also lower the degrees of freedom in the dataset, which can be beneficial for avoiding the curse of dimension. The curse of dimension refers to the problem of sparsity and lack of data points in high-dimensional spaces, which can lead to poor model performance and increase the risk of overfitting. Dimension reduction methods can mitigate this problem, resulting in a denser and more informative data representation. This can enhance the accuracy and robustness of machine learning models by providing them with a more representative and balanced dataset for training.

However, the effectiveness of dimension reduction is contingent upon multiple factors, including the dataset's structure, the specific methods of dimension reduction used, the number of retained lower dimensions, the choice of classifier, and the optimal parameter selection for each classifier. To achieve the best performance, it is crucial to consider these factors jointly and optimize the classification accuracy across them.

The main objective of this research project is to investigate the potential of dimension reduction in improving the accuracy of image classification using the Fashion-MNIST dataset. The study aims to address four key research questions:

- What is the most effective approach for revealing the underlying relationships among the 10 classes of images in the dataset, with the goal of constructing a low-dimensional "map" that preserves the dissimilarities between data points and the information of each class?

- Identifying the optimal classification models for the Fashion-MNIST dataset, both with and without the use of low-dimensional mapping, and comparing their respective performances.

- Assessing the impact of the number of dimensions used from the dimension reduction techniques on the performance of the classification models.

- Evaluating the training and testing time of different models, considering the use of low-dimensional mapping techniques in comparison to the original dataset.

2 Dataset

Zalando SE is a German multinational fashion company that specializes in e-commerce. The company was founded in Berlin in 2008 and has since grown to become one of Europe's largest online fashion retailers, offering a broad range of clothing, footwear, and accessories for men, women, and children.

In addition to its retail operations, Zalando is also involved in various technology initiatives related to fashion and e-commerce. One such initiative is the creation of the Fashion-MNIST dataset, which was developed by the company's research team in 2017 [1]. The Fashion-MNIST dataset is a collection of images of fashion items, such as clothing and footwear, designed to be a drop-in replacement for the classic MNIST handwritten digits dataset commonly used in machine learning research.

The Fashion-MNIST dataset consists of 70,000 grayscale images of 28x28 pixels, divided into 60,000 training images and 10,000 test images. Each image corresponds to one of 10 possible classes, representing different types of fashion items: 0: T-shirt/top, 1: Trouser, 2: Pullover, 3: Dress, 4: Coat, 5: Sandal, 6: Shirt, 7: Sneaker, 8: Bag, 9: Ankle boot.

The training data is 133 MB and the test data is 22.2 MB.

3 Methodology

In this study, four different dimension reduction techniques, namely linear Principal Component Analysis (PCA), kernel PCA, ISOMAP, and UMAP, are applied to the Fashion-MNIST dataset. Prior to applying each method, the dataset is preprocessed accordingly to ensure compatibility. It is important to note that the maximum number of lower dimensions mapping is set to 65, taking into consideration the processing capabilities of the computer used for this project.

Models are intended selected for comparison with two linear classifiers, two non-linear classifiers, and two ensemble methods classifiers, namely logistic regression, linear discriminant analysis (LDA), K-nearest neighbors (KNN), kernel Support Vector Machine (SVM), Random Forest, and Gradient Boosting Trees. A 10-fold cross-validation is used for identifying the optimal parameters for each model in conjunction with different dimension reduction methods and varying numbers of dimensions.

The sizes of the lower embedding dimensions range from 2 to 65. In order to lower the computational complexity, when to find the best parameters for each type of classifier, each dimension reduction method, and different numbers of dimensions, 10-fold cross-validation is performed at 10, 20, 30 until 65 dimensions. For example, the best k number for KNN classifier for 2-10 dimensions is searched from 10-fold cross-validation at 10 dimensions. For 11-20 dimensions, the best k is from the cross-validation for 20 dimensions. This allows for a comprehensive evaluation of the performance of each model at different levels of dimension reduction, capturing the effects of varying levels of data compression on classification accuracy.

Based on the best parameters, models are trained and applied to the 2000 test dataset. The test accuracy against sizes of the lower embedding dimensions is plotted for each classifier, each dimension reduction method. Based on the result, the top-performing classifiers and their corresponding number of lower embedding dimensions can be found.

The benchmark test accuracy is also calculated by using the original standard data set with 784 dimensions for each type of classifier. 10-fold cross-validation is still used here.

After determining the optimal classifiers on lower embedding dimensions through the previous steps, the training and testing times are recorded for each classification method. These times are then compared with the benchmark optimal models, which do not involve any dimension reduction. By comparing the training and testing times of the different models, the computational efficiency of each classifier in conjunction with the specific dimension reduction technique can be evaluated. This analysis provides insights into the trade-offs between classification accuracy and computational cost, helping to identify the most efficient approach for image classification with the Fashion-MNIST dataset.

4 Data Preprocessing

4.1 Train-Test Split

Due to the processing capability of the computer utilized in this project, downsampling is performed on the original training and testing dataset. By randomly selecting 1000 samples for each class in

the training dataset, the final training set is with 10000 images. Similarly, the test dataset is also downsampled to contain 200 images for each class, resulting in a total of 2000 test images.

4.2 Data Cloud Centering and Standardization

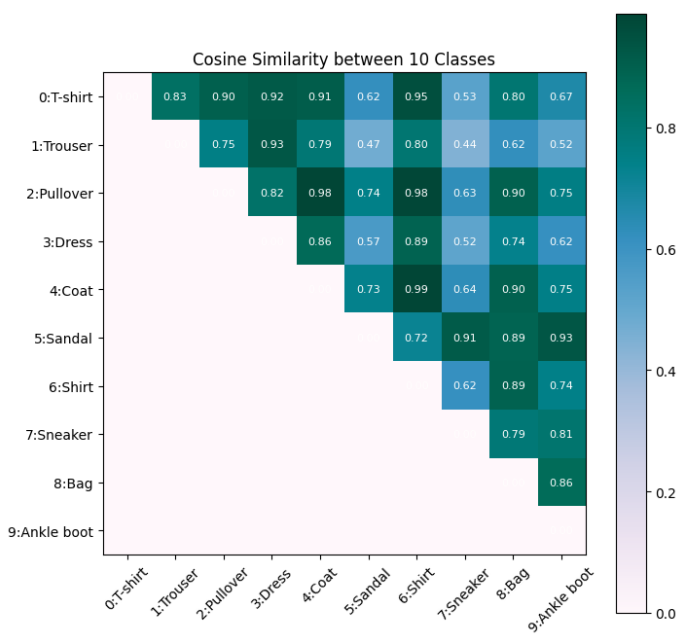
Linear PCA, as well as Kernel PCA, are known to be sensitive to the scale of features, and proper centering of data is critical for their effective implementation. In the case of grayscale pixel values in the range of 0 to 255, a simple division by 255 followed by centering is sufficient for both linear PCA and Kernel PCA. On the other hand, UMAP and ISOMAP are used to estimate the intrinsic geometry of a data manifold based on the neighborhood relationships of data points. They are both distance-based methods that rely on measuring distances between data points, so division by 255 is sufficient before applying these 2 dimension reduction techniques.

4.3 Challenge of Fashion-MNIST Dataset

As part of the exploratory data analysis (EDA), a visual examination of multiple images from the dataset is performed. Additionally, the pairwise cosine similarity between the 10 classes is calculated to evaluate potential challenges. This involves calculating the mean standard image vectors with a dimension of 784. Subsequently, the cosine similarity is computed between mean image vectors, denoted as $x \in R^{784},\ y \in R^{784}$:

$$S_{xy} = \frac{x^T y}{||x|| * ||y||}$$

The higher the S_{xy}, the more similar the two categories are. In particular, for class 0 (T-shirt/top), class 2 (Pullover), class 4 (Coat), and class 6 (Shirt), all the cosine similarities are found to be above 95%. These are really difficult classifications to make, even by human eye examination. This observation highlights the potential challenges in this classification project.



Some sample plots of classes 0, 2, 4 and 6 are shown below:

Class 0: T-shirt/top

Class 2: Pullover

Class 4: Coat

Class 6: Shirt

5 Results and Evaluation

5.1 Dimension Reduction

Dimension reduction algorithms tend to fall into two categories: those that seek to preserve the pairwise distance structure amongst all the data samples and those that favor the preservation of local distances over global distances. Algorithms such as PCA [4, 5], fall into the former category while t-SNE, ISOMAP [2], UMAP [3] all fall into the latter category.

5.1.1 Dimension Reduction for Linear PCA

Principal Component Analysis (PCA) is a popular unsupervised machine learning technique used for linear dimension reduction and feature extraction from high-dimensional data. It involves transforming the original features of a dataset into a lower-dimensional space while retaining as much of the original variance as possible. PCA identifies the principal components, which are linear combinations of the original features that capture the most important information in the data. Given a data set with m centered observations, $x^i \in R^d,\ s.t.\ i \in \{1, 2, ..., m\}$, PCA finds the

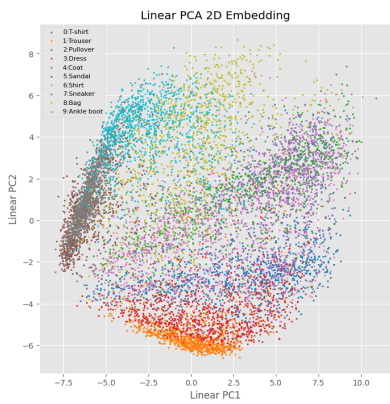
direction that maximizes the variance.

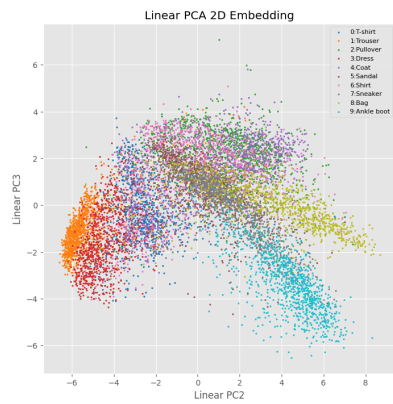
$$
\begin{aligned}
w^* &= argmax_{||w||\leq 1} \frac{1}{m} \sum_i (w^T x^i)^2 \\
&= argmax_{||w||\leq 1} \frac{1}{m} w^T X X^T w \\
s.t.\ X &= (x^1, x^2, ..., x^m)
\end{aligned}
$$

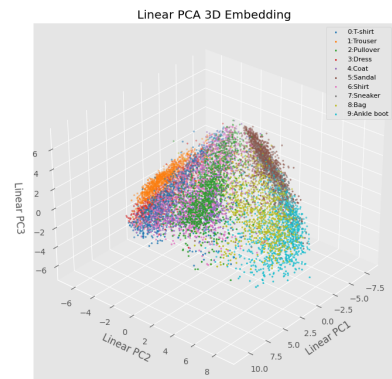
w^* can be found as the largest eigenvector by solving the eigenvalue problem:

$$
\begin{aligned}
C &= \frac{1}{m} X X^T \\
Cw &= \lambda w
\end{aligned}
$$

Plots of the 2D and 3D embedding of the 10000 training Fashion-MNIST dataset are shown below:







5.1.2 Dimension Reduction for Kernel PCA

The kernel PCA broadens the scope of PCA which expands the features by non-linear transformations, and then applies PCA in this transformed feature space. For linear PCA, the principal component lies in the span of the data and can be expressed as: $w = \sum_{i=1}^{m} \alpha_i x^i = X\alpha$.

$$
x^{iT} C w = \frac{1}{m} x^{iT} X X^T X \alpha = \lambda x^{iT} w = \lambda x^{iT} X \alpha
$$

In matrix form, it can be expressed as:

$$
\frac{1}{m} X^T X X^T X \alpha = \lambda X^T X \alpha
$$

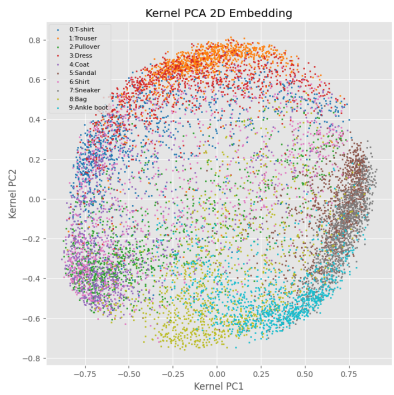
$X^T X$ can be replaced by kernel matrix K ("Gram matrix") which is a PSD matrix and suppose it is invertible, we have:

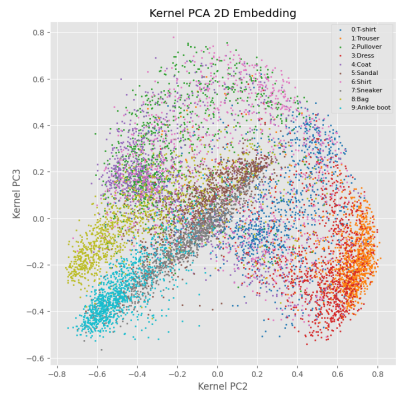
$$
\frac{1}{m} K \alpha = \lambda \alpha
$$

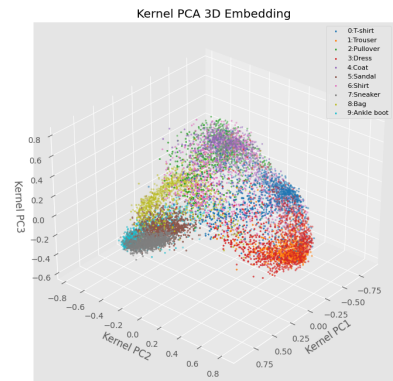
Based on this, we can find the best principle component by eigen-decomposition on K.

The choice of the kernel for kernel PCA is an important decision in the experiment. The cosine kernel, which measures similarity based on the cosine of the angle between feature vectors, is particularly suitable for this dataset where the angle between feature vectors carries a lot of information of the dissimilarities among the various classes. This is supported by the observation that the radial basis function kernel (RBF kernel) mapping is nearly identical to linear PCA, and considering the previous positive results from the cosine similarity analysis. As a result, the cosine kernel is chosen for the experiment.

$$\text{Cosine Kernel: } k(x, y) = \frac{x^T y}{||x|| * ||y||}$$







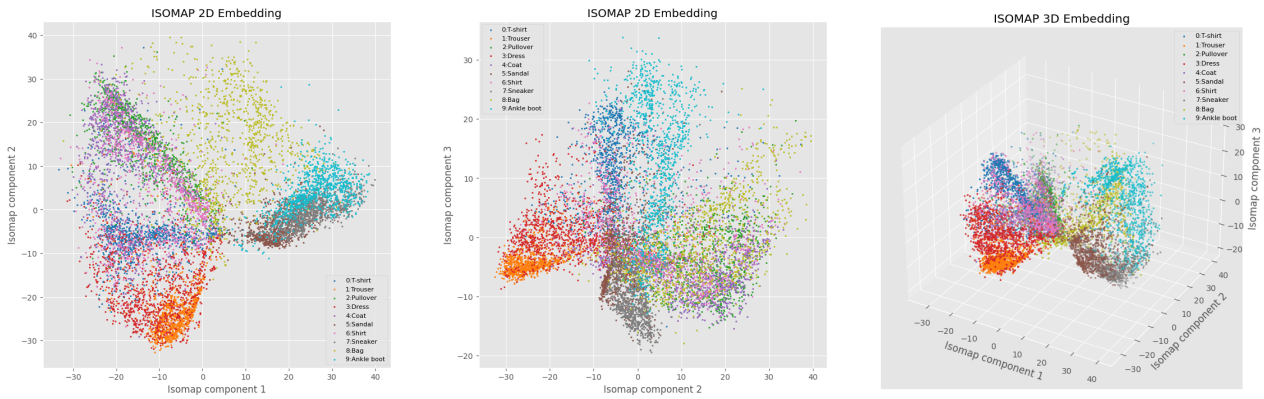
5.1.3 Dimension Reduction for ISOMAP

ISOMAP (Isometric Feature Mapping) is a non-linear dimension reduction technique that is suitable for data that lies on a low-dimensional manifold embedded in a high-dimensional space. It is particularly useful for data that has a nonlinear structure or exhibits complex, curved patterns. The key idea of ISOMAP is to produce low dimensional representation which preserves "walking-distance" over the data cloud (manifold) [2]. Suppose there are m data points:

- Find k neighbors N(i) for each data point x^i and build the adjacency matrix $A \in R^{m*m}$ with each neighbor's Euclidean distance
- Build the shortest path distance matrix $D \in R^{m*m}$ based on matrix A
- Based on the usage of matrix $H = I - \frac{1}{m}11^T$, calculate $G := -\frac{1}{2}HD^2H = U\Lambda U^T \approx Z^T Z$
- $Z^T = [u^1, u^2.., u^k] \begin{bmatrix} \lambda_1^{1/2} & & \\ & ... & \\ & & \lambda_k^{1/2} \end{bmatrix}$

Z^T is the new coordinate matrix of the original dataset with dimension m*k. Each row corresponds to a data point.

The choice of the number of neighbors k in ISOMAP is crucial, as it directly affects the quality of the final embedding. ISOMAP requires that the data points have sufficient neighborhood connectivity, allowing for the estimation of geodesic distances on the manifold. However, using too large a number of neighbors may result in the loss of larger-scale global patterns in the data, while too small a number of neighbors may result in disconnection in the manifold. Therefore, careful tuning of the number of neighbors in ISOMAP is necessary. For the Fashion-MNIST dataset, 5 neighbors result with a decent embedding.



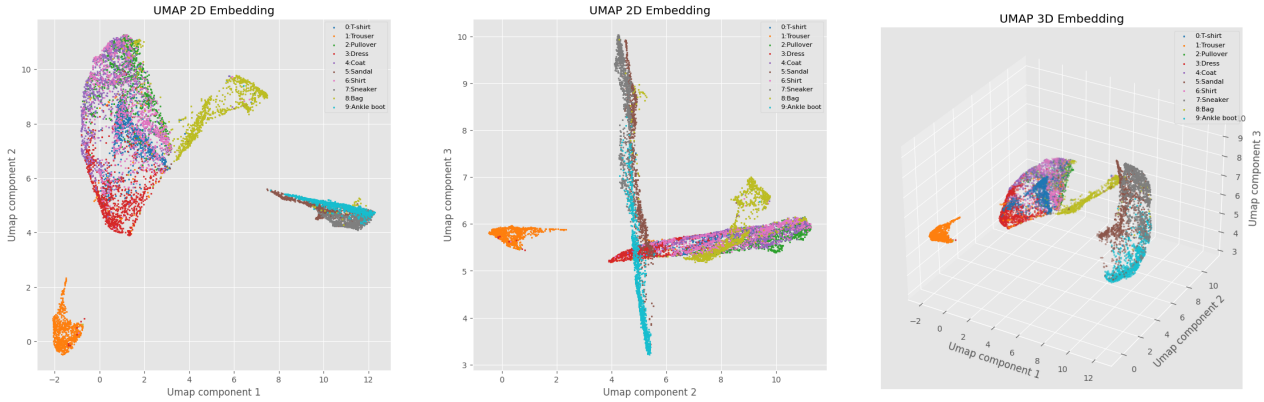
5.1.4 Dimension Reduction for UMAP

UMAP (Uniform Manifold Approximation and Projection) is another non-linear dimension reduction technique that this project explored. It is particularly popular nowadays for data visualization and exploration. UMAP is constructed from a theoretical framework based in Riemannian geometry and algebraic topology. The result is a practical scalable algorithm that is applicable to real-world data. The UMAP algorithm is competitive with t-SNE for visualization quality and arguably preserves more of the global structure with superior run-time performance. Furthermore, UMAP has no computational restrictions on the embedding dimension, making it viable as a general-purpose dimension reduction technique for machine learning [3].

UMAP, similar to ISOMAP, starts by constructing a pairwise distances matrix A. However, unlike ISOMAP, UMAP further constructs and optimizes a fuzzy simplicial set by minimizing a cross-entropy loss function. The optimization process involves fine-tuning the low-dimensional embedding to obtain an optimized result.

The detailed mathematical derivation will not be listed here but can be found in reference [3]. There are 2 important parameters that need to be tuned when forming UMAP embedding: *n_neighbors* and *min_dist*. Small values of *n_neighbors* will focus more on local structures and will result in a fine-grained cluster structure that may be more a result of patterns of noise in the data than actual clusters. *min_dist* determines how tightly the points are packed in the embedding space. A lower value of *min_dist* results in a more compact embedding, with points closer together, while a higher value leads to a more spread-out embedding. Too low of a value may cause points to overlap and become indistinguishable, while too high of a value may result in a loss of local structure. It is common to try different values of *min_dist* and visually inspect the embeddings to determine the optimal value for a given dataset.

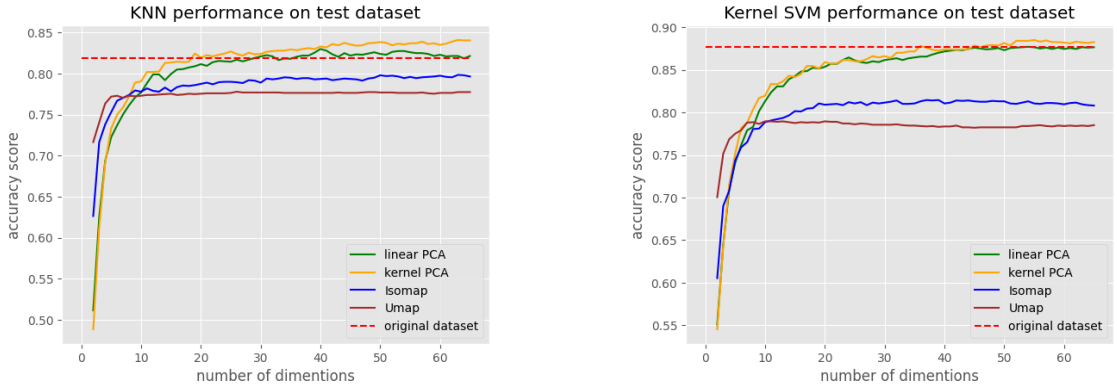
For the Fashion-MNIST dataset, the ideal parameter *n_neighbors* is set to be 15, and *min_dist* is set to be 0.2.



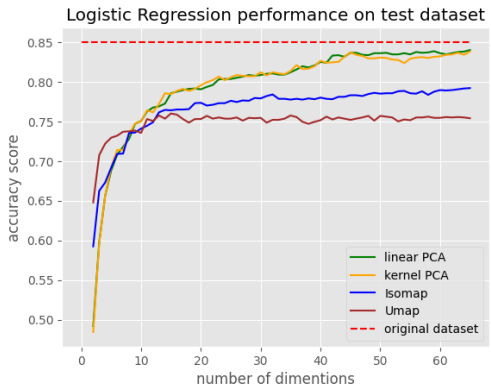
5.2 Classification versus the Number of Dimensions Used

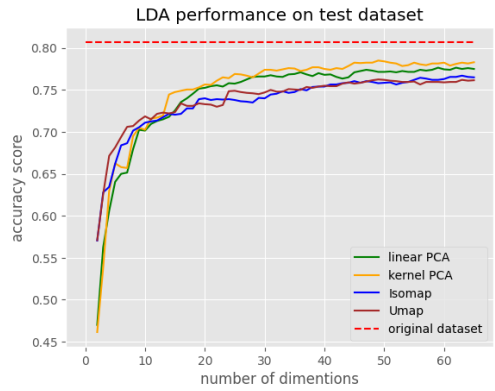
The results of previous sections indicate that UMAP and ISOMAP seem to be more effective in separating the 10 classes compared to PCA and kernel PCA on the first, second, and third dimensions. Furthermore, all four dimension reduction techniques struggle with classes 0, 2, 4, and 6, as evidenced by the overlapping of dots in pink, purple, dark blue, and green.

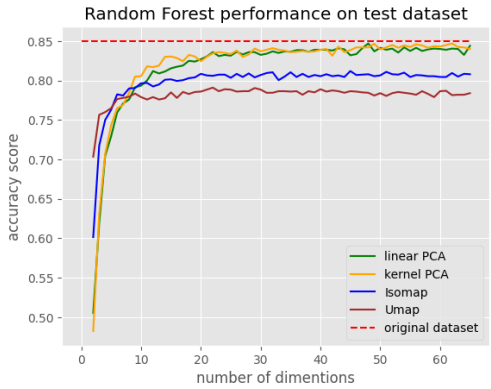
Interestingly, when applying the lower-dimensional data to different six classifiers, namely K-nearest neighbors (KNN), kernel Support Vector Machine (SVM), logistic regression, linear discriminant analysis (LDA), Random Forest, and Gradient Boosting Trees, other than LDA, UMAP shows limited performance after using 15 or more dimensions, while PCA and kernel PCA outperform both ISOMAP and UMAP with increasing dimensions used (all parameters of classifiers trained with certain numbers of lower dimension data is searched by 10-fold cross-validation).

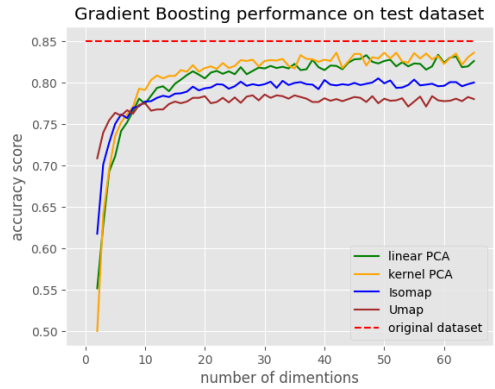


The comparison with the top-performing classifiers of each dimension reduction technique and its corresponding benchmark (the original standardized dataset with 784 dimensions) are shown below. Please note that the yellow portions indicate the models that outperform the benchmarks in terms of performance.









Model	Dimensionality Reduction Techniques	Number of Dimention Usage	Outperform benchmark with up to 65 dimentions	Test Accuracy	Training Time (s)	Training time saved compared with benchmark	Test Time (s)	Test time saved compared with benchmark
KNN	Original Dataset (benchmark)	784	-	0.819	0.009	-	0.645	-
	Linear PCA	40	Y	0.830	0.003	66.67%	0.399	38.14%
	Kernel PCA	63	Y	0.841	0.002	77.78%	0.499	22.64%
	Isomap	62	N	0.798	0.005	44.44%	0.546	15.35%
	Umap	26	N	0.778	0.006	33.33%	0.408	36.74%
Kernel SVM	Original Dataset (benchmark)	784	-	0.876	8.811	-	5.467	-
	Linear PCA	54	Y	0.877	1.480	83.20%	1.224	77.61%
	Kernel PCA	55	Y	0.885	1.266	85.63%	1.350	75.31%
	Isomap	37	N	0.811	1.230	86.04%	1.522	72.16%
	Umap	10	N	0.790	3.678	58.26%	0.781	85.71%
Logistic Regression	Original Dataset (benchmark)	784	-	0.851	7.059	-	0.002	-
	Linear PCA	65	N	0.840	2.823	60.01%	0.001	50.00%
	Kernel PCA	65	N	0.838	2.132	69.80%	0.002	0.00%
	Isomap	65	N	0.792	3.978	43.65%	0.001	50.00%
	Umap	15	N	0.768	21.026	-197.86%	0.001	50.00%
LDA	Original Dataset (benchmark)	784	-	0.808	1.375	-	0.005	-
	Linear PCA	59	N	0.782	0.078	94.33%	0.001	80.00%
	Kernel PCA	49	N	0.785	0.063	95.42%	0.001	80.00%
	Isomap	63	N	0.767	0.077	94.40%	0.001	80.00%
	Umap	49	N	0.763	0.060	95.64%	0.001	80.00%
Random Forest	Original Dataset (benchmark)	784	-	0.853	23.840	-	0.183	-
	Linear PCA	48	N	0.847	13.685	42.60%	0.130	28.96%
	Kernel PCA	63	N	0.847	13.480	43.46%	0.110	39.89%
	Isomap	45	N	0.812	11.432	52.05%	0.114	37.70%
	Umap	22	N	0.791	7.743	67.52%	0.117	36.07%
Gradient Boosting Trees	Original Dataset (benchmark)	784	-	0.854	17.859	-	0.034	-
	Linear PCA	59	N	0.834	14.107	21.01%	0.024	29.41%
	Kernel PCA	42	N	0.836	12.847	28.06%	0.031	8.82%
	Isomap	49	N	0.805	10.174	43.03%	0.025	26.47%
	Umap	30	N	0.786	6.344	64.48%	0.013	61.76%

Below, we provide a comprehensive analysis of each top-performing classification method's model that utilizes dimension reduction techniques compared to the benchmark.

KNN - kernel PCA (d=63)			
class	precision	recall	f1-score
0	0.75	0.82	0.79
1	0.99	0.97	0.98
2	0.74	0.73	0.74
3	0.87	0.86	0.86
4	0.71	0.80	0.75
5	0.95	0.86	0.91
6	0.65	0.52	0.57
7	0.85	0.92	0.88
8	0.96	0.97	0.97
9	0.93	0.94	0.94
total test accuracy=0.841			

KNN- original standardized data (d=784)			
class	precision	recall	f1-score
0	0.73	0.82	0.78
1	0.99	0.97	0.98
2	0.68	0.72	0.70
3	0.90	0.83	0.86
4	0.69	0.73	0.71
5	0.99	0.69	0.81
6	0.59	0.53	0.56
7	0.80	0.93	0.86
8	0.97	0.95	0.96
9	0.83	0.94	0.88
total test accuracy=0.819			

Kernel SVM - kernel PCA (d=55)			
class	precision	recall	f1-score
0	0.82	0.85	0.84
1	0.99	0.97	0.98
2	0.81	0.84	0.83
3	0.87	0.90	0.89
4	0.81	0.79	0.80
5	0.97	0.96	0.96
6	0.74	0.68	0.71
7	0.90	0.94	0.92
8	0.96	0.99	0.97
9	0.96	0.93	0.95
total test accuracy=0.885			

Kernel SVM - original standardized data (d=784)			
class	precision	recall	f1-score
0	0.80	0.83	0.82
1	1.00	0.97	0.98
2	0.80	0.80	0.80
3	0.87	0.90	0.89
4	0.80	0.81	0.80
5	0.95	0.95	0.95
6	0.70	0.67	0.68
7	0.90	0.94	0.92
8	0.98	0.98	0.98
9	0.97	0.93	0.95
total test accuracy=0.876			

Logistic Regression - linear PCA (d=65)			
class	precision	recall	f1-score
0	0.80	0.78	0.79
1	0.99	0.95	0.97
2	0.76	0.69	0.72
3	0.80	0.89	0.84
4	0.69	0.80	0.74
5	0.95	0.93	0.94
6	0.61	0.57	0.59
7	0.89	0.92	0.90
8	0.95	0.94	0.95
9	0.95	0.94	0.95
total test accuracy=0.84			

Logistic Regression - original standardized data (d=784)			
class	precision	recall	f1-score
0	0.79	0.79	0.79
1	0.99	0.96	0.98
2	0.77	0.73	0.75
3	0.83	0.88	0.86
4	0.71	0.83	0.77
5	0.93	0.93	0.93
6	0.68	0.57	0.62
7	0.88	0.93	0.90
8	0.96	0.95	0.95
9	0.96	0.94	0.95
total test accuracy=0.851			

LDA - kernel PCA (d=49)			
class	precision	recall	f1-score
0	0.76	0.73	0.74
1	0.99	0.93	0.96
2	0.70	0.62	0.66
3	0.76	0.84	0.80
4	0.64	0.76	0.70
5	0.85	0.86	0.86
6	0.53	0.49	0.51
7	0.81	0.77	0.79
8	0.93	0.94	0.94
9	0.87	0.91	0.89
total test accuracy=0.785			

LDA - original standardized data (d=784)			
class	precision	recall	f1-score
0	0.77	0.75	0.76
1	1.00	0.93	0.96
2	0.75	0.68	0.71
3	0.80	0.85	0.82
4	0.67	0.80	0.73
5	0.85	0.86	0.86
6	0.58	0.55	0.56
7	0.83	0.88	0.86
8	0.93	0.91	0.92
9	0.92	0.87	0.89
total test accuracy=0.808			

Random Forest - linear PCA (d=48)			
class	precision	recall	f1-score
0	0.80	0.82	0.81
1	1.00	0.96	0.98
2	0.76	0.76	0.76
3	0.82	0.86	0.84
4	0.74	0.80	0.77
5	0.91	0.91	0.91
6	0.71	0.57	0.63
7	0.88	0.88	0.88
8	0.93	0.96	0.95
9	0.90	0.94	0.92
total test accuracy=0.847			

Random Forest - original standardized data (d=784)			
class	precision	recall	f1-score
0	0.79	0.79	0.79
1	0.99	0.95	0.97
2	0.78	0.76	0.77
3	0.84	0.9	0.87
4	0.74	0.84	0.79
5	0.95	0.93	0.94
6	0.68	0.55	0.6
7	0.88	0.89	0.89
8	0.93	0.98	0.96
9	0.94	0.94	0.94
total test accuracy=0.853			

Gradient Boosting Trees - kernel PCA (d=42)			
class	precision	recall	f1-score
0	0.78	0.82	0.80
1	1.00	0.94	0.97
2	0.73	0.73	0.73
3	0.84	0.86	0.85
4	0.73	0.77	0.75
5	0.90	0.94	0.92
6	0.59	0.53	0.56
7	0.89	0.85	0.87
8	0.94	0.96	0.95
9	0.91	0.93	0.92
total test accuracy=0.836			

Gradient Boosting Trees - original standardized data (d=784)			
class	precision	recall	f1-score
0	0.79	0.80	0.79
1	1.00	0.96	0.98
2	0.78	0.77	0.78
3	0.84	0.89	0.86
4	0.76	0.81	0.79
5	0.96	0.93	0.94
6	0.65	0.58	0.61
7	0.90	0.91	0.90
8	0.95	0.98	0.97
9	0.93	0.94	0.93
total test accuracy=0.854			

The experimental results show that the kernel SVM with data from kernel PCA with 55 dimensions outperforms all other classifiers in terms of classification accuracy. Furthermore, compared to its benchmark model which is also the optimal classification model among all models trained with the original standardized data, the kernel SVM with kernel PCA reduced the training time by 85%.

The Random Forest and Gradient Boosting Trees also demonstrated competitive performance (RF: above 84%; GBT: above 83% for both liner PCA and kernel PCA data), albeit with longer training times. However, dimension reduction was able to significantly reduce the training time for these ensemble methods by more than 40% for RF and 20% for GBT.

On the other hand, the LDA classifier showed the poorest performance, with accuracy below 80% for all dimension reduction techniques, in spite of the shortest training time. LDA is a parametric method, meaning it makes certain assumptions about the underlying data distribution. It is typically suitable for data that with multivariate normal distribution within each group, and linear

relationships between features. One possible reason for the poor performance might be the underlying data distribution not meeting the assumptions of LDA. Insufficient low-dimension embedding might be the other reason. Based on the plot of LDA performance, all classifiers' accuracy keeps increasing until dimension 65. If add more lower dimensions, LDA might result in a compatible performance.

Regarding the utilization of dimension reduction techniques, it is generally expected that increasing the number of dimensions to a certain point would improve accuracy for all classifiers. Afterward, noises are added in by more dimensions and will result in decreased performance. However, UMAP and ISOMAP exhibit an exception to this trend. It is observed that UMAP consistently outperforms all other dimension reduction methods across various classifiers when using fewer than 10 dimensions. This further validates UMAP's exceptional ability for data visualization. However, UMAP shows limited effectiveness in improving accuracy when adding more dimensions beyond 10, as there is no significant increase in accuracy for classifiers such as logistic regression, KNN, kernel SVM, etc. This phenomenon is also observed in ISOMAP.

6 Conclusion and Future Work

After conducting an in-depth investigation into dimension reduction techniques and various classification models on the Fashion-MNIST dataset, the findings highlight the importance of jointly selecting appropriate classifiers and dimension reduction techniques and also show the potential benefits of using dimension reduction methods in improving the efficiency and effectiveness of classification models. There still remains lots of area for future research. Further investigations can be carried out to explore different dimension reduction methods, such as t-SNE, Laplacian eigenmaps, Sammon mapping, and their impact on classification performance.

Furthermore, due to limitations in the computational capabilities of the computer utilized in this project, the maximum lower dimension embedding was set at 65. However, it may be worthwhile to explore lower-dimension embeddings of up to 150 in future research. Based on observations, increasing the number of lower dimensions may allow for more information to be added to the dataset, potentially resulting in more classifiers outperforming the benchmarks. Nevertheless, it is important to acknowledge that increasing the number of dimensions also raises the risk of overfitting.

Finally, it is worth mentioning that among the various research conducted on the Fashion-MNIST dataset, the convolutional neural network (CNN) has emerged as a popular and highly accurate classifier, capable of achieving accuracy levels of up to 98% when using the standardized dataset [6]. Exploring the combination of dimension reduction techniques, such as PCA or UMAP, with CNN could be an intriguing avenue for future research which might provide valuable insights into the potential benefits and trade-offs of utilizing dimension reduction in conjunction with CNN for image classification tasks.

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